A Worked Tutorial on Digit-Recurrence Algorithms for Division and Square Root

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Abstract

Digit-recurrence algorithms, especially those in the SRT (Sweeney, Robertson, and Tocher) family, remain a standard approach to high-speed hardware division and square root. They generate one output digit per iteration while maintaining a scaled residual inside a bounded interval. Their central hardware advantage is redundancy: signed digit sets such as $\{-1, 0, 1\}$ or $\{-2, -1, 0, 1, 2\}$ create overlap between adjacent digit-selection regions, so the next digit can be chosen from truncated estimates of the residual, divisor, or partial root rather than from an exact full-width carry-propagated value.

This tutorial develops division and square root in parallel. For each operation it derives the result expansion, the scaled residual recurrence, the containment condition, the exact digit-selection intervals, and practical selector tables built from truncated state. Worked radix-2 and radix-4 examples connect the algebra to concrete hardware logic, and a final section shows how deterministic overlap resolution can simplify the selector.

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1 Introduction

Division and square root are fundamentally iterative in hardware. Unlike addition, which can often be completed in a shallow constant-depth network, or multiplication, whose partial-product reduction is mostly feed-forward, division and square root repeatedly update an internal state. Each iteration must do two things:

1. choose the next result digit, and
2. update the residual (partial remainder).

The latency of the unit is therefore dominated by this per-iteration loop.

The central difficulty is digit selection. With a nonredundant digit set, the selector would have to know the residual accurately enough to distinguish between adjacent regions with essentially no tolerance for uncertainty. That requirement usually pulls a full carry-propagating addition into the critical path. SRT-style digit-recurrence algorithms avoid that cost by using a redundant signed-digit set. Redundancy creates overlap between neighboring valid regions. Inside such an overlap, more than one digit is acceptable, so the selector may safely use truncated estimates of the residual and of the divisor or partial root. That slack is what makes a carry-save residual practical.

Two tutorial conventions are worth stating up front. First, we intentionally normalize operands more tightly than the most general SRT theory requires. For division we use

$$N \in \left(0, \frac{1}{4}\right), \qquad D \in \left[\frac{1}{2}, 1\right) \text{ or } D \in \left[-1, -\frac{1}{2}\right),$$

and for square root we use

$$X \in \left[\frac{1}{4}, 1\right).$$

These choices are pedagogical: they keep the startup step simple, so the tutorial can focus on the recurrence and selector construction rather than on prescaling details. When $D < 0$, the quotient sign is absorbed directly into the signed-digit expansion, so the later negative-divisor selector tables are not a different algorithm but the reflected version of the same construction. Second, whenever we convert a truncated fixed-point word into an integer table index, we write $\mathrm{Int}(\cdot)$ for "remove the binary point and interpret the result as an integer."

The rest of the tutorial is organized as follows. Section 2 introduces notation, digit sets, and the redundancy factor. Section 3 develops division from first principles and works through radix-2 and radix-4 selectors. Section 4 repeats the same program for square root, highlighting the additional dependence on the evolving partial root. Section 5 explains how overlap can be resolved deterministically to simplify the hardware selector, and section 6 summarizes the main design lessons.

2 Preliminaries and Tutorial Conventions

2.1 Notation used throughout

Table 1: Symbols used repeatedly in the tutorial.

Symbol	Meaning
B	Radix of the digit-recurrence algorithm. In the worked examples, $B \in \{2, 4\}$.
a	Maximum digit magnitude in the symmetric digit set $\{-a, -a+1, \ldots, a-1, a\}$.
ρ	Redundancy factor, defined by $\rho = a/(B-1)$. Overlap exists when $\rho > 1/2$.
Q_j, q_{j+1}	Quotient approximation after j selected digits, and the next quotient digit.
S_j, s_{j+1}	Partial-root approximation after j selected digits, and the next root digit.
R_j	Scaled residual (partial remainder) at iteration j.
N, D, X	Dividend, divisor, and radicand.
t_j	Truncated estimate of the shifted residual used by the digit selector.
τ_j	Integerized version of the truncated residual estimate.
δ	Integerized truncated divisor estimate used by the quotient selector.
σ_j	Integerized truncated partial-root estimate used by the root selector.

2.2 Redundant signed-digit sets

For both division and square root we choose each new digit from a symmetric signed-digit set

$$d_{j+1} \in \{-a, -a+1, \ldots, a-1, a\}, \tag{2.1}$$

where d_{j+1} stands for either q_{j+1} or s_{j+1} depending on the context. The corresponding redundancy factor is

$$\rho = \frac{a}{B-1}. \tag{2.2}$$

The basic design rule is

$$\rho > \frac{1}{2}. \tag{2.3}$$

To see why this matters, suppose the ideal selector says that digit k is valid whenever the residual lies in an interval of the form

$$L_k \leq \text{residual} \leq U_k,$$

and digit $k+1$ is valid whenever

$$L_{k+1} \leq \text{residual} \leq U_{k+1}.$$

When $\rho > 1/2$, these adjacent intervals overlap:

$$U_k > L_{k+1}.$$

That overlap is exactly the slack that allows truncated selection. If $\rho = 1/2$, the intervals merely touch; if $\rho < 1/2$, they separate and the selector must know the residual more precisely.

In practice one usually chooses

$$\frac{1}{2} < \rho \leq 1, \tag{2.4}$$

which is equivalent to

$$\frac{B-1}{2} < a \leq B-1. \tag{2.5}$$

The upper end $\rho = 1$ is maximally redundant; values closer to $1/2$ are minimally redundant. More redundancy increases overlap but also enlarges the set of digit multiples that must be formed in the datapath.

2.3 Why the tutorial normalizations are slightly tighter

The SRT framework itself is more general than the specific normalizations used here. In a floating-point implementation, exponents are handled separately and the significands are then scaled into a convenient interval. This tutorial assumes the already-normalized cases

$$N \in \left(0, \frac{1}{4}\right), \qquad D \in \left[\frac{1}{2}, 1\right) \text{ or } D \in \left[-1, -\frac{1}{2}\right), \tag{2.6}$$

$$X \in \left[\frac{1}{4}, 1\right). \tag{2.7}$$

These bounds imply

$$0 < \left|\frac{N}{D}\right| < \frac{1}{2},$$

and

$$\sqrt{X} \in \left[\frac{1}{2}, 1\right).$$

As a result, division can start from $Q_0 = 0$ and square root can start from $S_0 = 1$. The payoff is a cleaner presentation: the initialization algebra stays transparent, the startup cases stay small, and the tutorial can spend its time on the recurrence itself rather than on prescaling machinery.

2.4 A three-step recipe for building a practical selector

Each worked example in sections 3 and 4 follows the same pattern.

1. Derive the exact residual recurrence and the exact interval that makes a candidate digit valid.
2. Replace the exact residual (and the divisor or partial root) with truncated estimates. The resulting uncertainty must still fit inside the overlap between neighboring exact intervals.
3. Convert the truncated fixed-point quantities into small integers τ_j, δ, and σ_j, and write the final selector logic as integer inequalities.

The continuity condition introduced later is the formal statement of Step 2. It guarantees that after truncation there is still at least one valid digit in every state box of the selector.

3 Division by Digit Recurrence

We begin from the exact relation

$$N = DQ_n + r, \tag{3.1}$$

where N is the dividend, D the divisor, Q_n the final quotient, and r the final *unscaled* remainder.

3.1 Quotient expansion

Let Q_j denote the quotient after j selected digits. The next digit update is

$$Q_{j+1} = Q_j + q_{j+1}B^{-(j+1)}. \tag{3.2}$$

Repeated substitution gives

$$Q_{j+1} = Q_0 + \sum_{i=0}^{j} q_{i+1}B^{-(i+1)}, \tag{3.3}$$

and therefore

$$Q_n = Q_0 + \sum_{i=0}^{n-1} q_{i+1}B^{-(i+1)}. \tag{3.4}$$

3.2 Approximation error and containment

Define the quotient error after $j+1$ selected digits by

$$\varepsilon_{j+1} = \frac{N}{D} - Q_{j+1}. \tag{3.5}$$

Substituting eq. (3.3) into the exact radix expansion of the true quotient shows that the error is the tail of the digit series:

$$\varepsilon_{j+1} = \sum_{i=j+1}^{\infty} q_{i+1}B^{-(i+1)}. \tag{3.6}$$

Because every digit satisfies $|q_{i+1}| \le a$,

$$\begin{aligned}
|\varepsilon_{j+1}| &= \left| \sum_{i=j+1}^{\infty} q_{i+1}B^{-(i+1)} \right| \\
&\le \sum_{i=j+1}^{\infty} aB^{-(i+1)} \\
&= aB^{-(j+2)} \sum_{k=0}^{\infty} B^{-k} \\
&= aB^{-(j+2)} \cdot \frac{1}{1-B^{-1}} \\
&= \frac{a}{B-1} B^{-(j+1)} \\
&= \rho B^{-(j+1)}.
\end{aligned} \tag{3.7}$$

Thus

$$\left| \frac{N}{D} - Q_{j+1} \right| \le \rho B^{-(j+1)}. \tag{3.8}$$

Multiplying by the positive quantity $B^{j+1}|D|$ gives the scaled form used in hardware:

$$\left|B^{j+1}(N - DQ_{j+1})\right| \leq \rho|D|. \tag{3.9}$$

This motivates the scaled residual

$$R_{j+1} = B^{j+1}(N - DQ_{j+1}), \tag{3.10}$$

for which the containment condition is simply

$$|R_{j+1}| \leq \rho|D|. \tag{3.11}$$

At the end of the algorithm,

$$R_n = B^n(N - DQ_n) = B^n r. \tag{3.12}$$

So R_n is the ordinary remainder scaled by B^n.

3.3 Residual recurrence

Starting from the definition of R_{j+1} and substituting the quotient update eq. (3.2),

$$\begin{aligned} R_{j+1} &= B^{j+1}(N - DQ_{j+1}) \\ &= B^{j+1}\left(N - D\left(Q_j + q_{j+1}B^{-(j+1)}\right)\right) \\ &= B^{j+1}(N - DQ_j) - q_{j+1}D \\ &= BR_j - q_{j+1}D. \end{aligned} \tag{3.13}$$

This is the standard digit-recurrence equation for division. Every iteration chooses q_{j+1} and then updates the residual with one multiply-by-radix and one subtract/add of a divisor multiple.

3.4 Ideal digit-selection intervals

3.4.1 Positive divisor

When $D \in [1/2, 1)$, the containment condition eq. (3.11) becomes

$$-\rho D \leq R_{j+1} \leq \rho D. \tag{3.14}$$

Substituting the recurrence eq. (3.13) yields

$$-\rho D \leq BR_j - q_{j+1}D \leq \rho D. \tag{3.15}$$

Now add $q_{j+1}D$ to all three parts:

$$(q_{j+1} - \rho)D \leq BR_j \leq (q_{j+1} + \rho)D. \tag{3.16}$$

So, for a fixed divisor D, digit q_{j+1} is valid precisely when

$$BR_j \in \left[(q_{j+1} - \rho)D,\ (q_{j+1} + \rho)D\right].$$

Because $\rho > 1/2$, neighboring intervals overlap. Indeed, the upper end of digit $k - 1$ equals $(k - 1 + \rho)D$, while the lower end of digit k equals $(k - \rho)D$, and

$$(k - 1 + \rho)D - (k - \rho)D = (2\rho - 1)D > 0.$$

That overlap is the slack the practical quotient selector will later consume.

3.4.2 Negative divisor

When $D \in [-1, -1/2)$, we have $|D| = -D$, so containment becomes

$$\rho D \le R_{j+1} \le -\rho D. \tag{3.17}$$

Substituting the recurrence gives

$$\rho D \le BR_j - q_{j+1}D \le -\rho D. \tag{3.18}$$

Adding $q_{j+1}D$ to all three terms gives

$$(q_{j+1} + \rho)D \le BR_j \le (q_{j+1} - \rho)D. \tag{3.19}$$

The interval endpoints appear "reversed" only because $D < 0$. Equivalently, digit q_{j+1} is valid when

$$BR_j \in \big[(q_{j+1} + \rho)D,\ (q_{j+1} - \rho)D\big].$$

The same overlap calculation still works because multiplying by a negative D swaps the ordering but not the existence of overlap.

3.5 Initialization

The tutorial normalization makes the startup step unusually clean. For positive divisors,

$$0 < \frac{N}{D} < \frac{1}{2}. \tag{3.20}$$

Hence it is natural to start from

$$Q_0 = 0. \tag{3.21}$$

The first update is

$$Q_1 = Q_0 + q_1 B^{-1} = q_1 B^{-1}. \tag{3.22}$$

The general error bound eq. (3.8) with $j = 0$ gives

$$\left| \frac{N}{D} - Q_1 \right| \le \rho B^{-1}. \tag{3.23}$$

Expanding the absolute value,

$$Q_1 - \rho B^{-1} \le \frac{N}{D} \le Q_1 + \rho B^{-1}. \tag{3.24}$$

Since $Q_1 = q_1 B^{-1}$, digit q_1 covers the interval

$$(q_1 - \rho)\, B^{-1} \le \frac{N}{D} \le (q_1 + \rho)\, B^{-1}. \tag{3.25}$$

Because the true quotient lies in $(0, 1/2)$, we may restrict the first digit to the nonnegative half of the digit set:

$$q_1 \in \{0, 1, \ldots, a\}.$$

That is exactly why the chosen normalization is convenient.

It is easy to verify coverage of $(0, 1/2)$. The leftmost possible first-step interval is obtained from $q_1 = 0$:

$$-\rho B^{-1} \le \frac{N}{D} \le \rho B^{-1}.$$

The rightmost interval is obtained from $q_1 = a$:

$$(a - \rho)B^{-1} \le \frac{N}{D} \le (a + \rho)B^{-1}.$$

Since $a = \rho(B - 1)$, the right endpoint becomes

$$(a + \rho)B^{-1} = \rho.$$

Because $\rho > 1/2$, the right edge reaches beyond $1/2$, and the first digit always exists.

For negative divisors, the same startup argument may be applied to $|D|$ and then reflected through the signed-digit tables of sections 3.7.2 and 3.8.2. Equivalently, one may view the negative-divisor case as the positive-divisor case followed by a quotient-sign reflection in the selected digits.

For the two radices used later, the startup intervals are especially simple.

Radix-2. Here $B = 2$, $a = 1$, and $\rho = 1$. Thus

$$\begin{aligned} q_1 = 0 &\Longrightarrow -\frac{1}{2} \le \frac{N}{D} \le \frac{1}{2}, \\ q_1 = 1 &\Longrightarrow 0 \le \frac{N}{D} \le 1. \end{aligned}$$

Multiplying by a positive divisor gives

$$\begin{aligned} q_1 = 0 &\Longrightarrow -\frac{1}{2}D \le N \le \frac{1}{2}D, \\ q_1 = 1 &\Longrightarrow 0 \le N \le D. \end{aligned}$$

Radix-4. Here $B = 4$, $a = 2$, and $\rho = 2/3$. Thus

$$\begin{aligned} q_1 = 0 &\Longrightarrow -\frac{1}{6} \le \frac{N}{D} \le \frac{1}{6}, \\ q_1 = 1 &\Longrightarrow \frac{1}{12} \le \frac{N}{D} \le \frac{5}{12}, \\ q_1 = 2 &\Longrightarrow \frac{1}{3} \le \frac{N}{D} \le \frac{2}{3}. \end{aligned}$$

Multiplying by a positive divisor gives

$$\begin{aligned} q_1 = 0 &\Longrightarrow -\frac{1}{6}D \le N \le \frac{1}{6}D, \\ q_1 = 1 &\Longrightarrow \frac{1}{12}D \le N \le \frac{5}{12}D, \\ q_1 = 2 &\Longrightarrow \frac{1}{3}D \le N \le \frac{2}{3}D. \end{aligned}$$

3.6 From exact intervals to a practical quotient selector

The intervals in section 3.4 are exact: they are written in terms of the true values of BR_j and D. A hardware implementation rarely has those values available in that form. Instead, the residual is typically kept in carry-save format, and the selector sees only a short carry-propagated prefix together with a truncated view of the divisor. This subsection introduces the generic truncation model that will be instantiated in the radix-2 and radix-4 examples.

Assume that the shifted residual BR_j is represented as the sum of a carry-save pair,

$$BR_j = Y_j + Z_j, \tag{3.26}$$

where Y_j and Z_j are fixed-point words. A short CPA adds only the first α fractional bits and produces a truncated value t_j. The truncation is one-sided: discarding the low-order tail makes the retained prefix no larger than the exact carry-propagated value. Therefore we can write

$$0 \le BR_j - t_j < 2^{-(\alpha-1)}. \tag{3.27}$$

Similarly, if the selector keeps only the first β fractional bits of the divisor, then the truncated divisor index δ satisfies

$$\delta \le 2^{\beta} D < \delta + 1. \tag{3.28}$$

The selector must still be correct for every exact pair (BR_j, D) consistent with the indices (t_j, δ). Consider the boundary between two adjacent digits $k-1$ and k for a positive divisor. From eq. (3.16), the upper exact boundary of digit $k-1$ is

$$U_{k-1}(D) = (k - 1 + \rho)D,$$

and the lower exact boundary of digit k is

$$L_k(D) = (k - \rho)D.$$

After truncation, a sufficient condition for the two selector regions still to overlap is

$$\min\bigl(U_{k-1,\mathrm{trunc}}\bigr) \ge \max\bigl(L_{k,\mathrm{trunc}}\bigr). \tag{3.29}$$

The minimum of the upper boundary uses the smallest divisor value in the truncated bin and subtracts one full residual-truncation error:

$$\min\bigl(U_{k-1,\mathrm{trunc}}\bigr) = (k - 1 + \rho)\, 2^{-\beta}\delta - 2^{-(\alpha-1)}, \tag{3.30}$$

provided the coefficient of D is positive. Likewise,

$$\max\bigl(L_{k,\mathrm{trunc}}\bigr) = (k - \rho)\, 2^{-\beta}(\delta + 1), \tag{3.31}$$

again for a positive coefficient of D. If the coefficient of D is negative, the roles of δ and $\delta + 1$ swap because increasing D moves the boundary in the opposite direction.

The selector-design problem now separates into two concrete tasks:

1. choose enough residual precision (the parameter α) and enough divisor/root precision (the parameter β), and
2. once those precisions are fixed, map the integerized state variables into digit regions.

The worked examples below carry out both steps explicitly.

3.7 Worked example: radix-2 division

For radix 2 we choose the maximally redundant digit set

$$q_{j+1} \in \{-1, 0, 1\},$$

so $a = 1$ and therefore

$$\rho = \frac{a}{B-1} = \frac{1}{2-1} = 1. \tag{3.32}$$

3.7.1 Positive divisor

Substituting $B = 2$ and $\rho = 1$ into eq. (3.16) gives the exact validity intervals

$$q_{j+1} = \begin{cases} 1 & \text{if } 0 \le 2R_j \le 2D, \\ 0 & \text{if } -D \le 2R_j \le D, \\ -1 & \text{if } -2D \le 2R_j \le 0. \end{cases} \tag{3.33}$$

Because the digit set is redundant, these intervals overlap:

$$q_{j+1} = 1 \text{ and } q_{j+1} = 0 \text{ are both valid when } 0 \le 2R_j \le D,$$

and similarly

$$q_{j+1} = 0 \text{ and } q_{j+1} = -1 \text{ are both valid when } -D \le 2R_j \le 0.$$

Let the truncated residual satisfy

$$0 \le 2R_j - t_j < 2^{-(\alpha-1)}. \tag{3.34}$$

Define the integer selector variables

$$\tau_j = \mathrm{Int}\big(2^{\alpha} t_j\big), \qquad \tau_j \le 2^{\alpha+1} R_j < \tau_j + 2, \tag{3.35}$$

$$\delta = \mathrm{Int}\big(2^{\beta} D_{\mathrm{trunc}}\big), \qquad \delta \le 2^{\beta} D < \delta + 1. \tag{3.36}$$

Since $D \in [1/2, 1)$, the divisor index ranges over

$$\delta \in \{2^{\beta-1}, 2^{\beta-1}+1, \ldots, 2^{\beta}-2, 2^{\beta}-1\}. \tag{3.37}$$

We now derive the continuity condition explicitly. There are only two neighboring boundaries to check.

Boundary between $q = 1$ and $q = 0$. Here

$$U_0(D) = (0+\rho)D = D, \qquad L_1(D) = (1-\rho)D = 0.$$

Using eqs. (3.30) and (3.31),

$$\begin{aligned} \min\big(U_{0,\mathrm{trunc}}\big) &= 2^{-\beta}\delta - 2^{-(\alpha-1)}, \\ \max\big(L_{1,\mathrm{trunc}}\big) &= 0. \end{aligned} \tag{3.38}$$

Therefore the overlap survives truncation if

$$2^{-\beta}\delta - 2^{-(\alpha-1)} \ge 0. \tag{3.39}$$

Boundary between $q = 0$ and $q = -1$. Here

$$U_{-1}(D) = (-1 + \rho)D = 0, \qquad L_0(D) = (0 - \rho)D = -D.$$

Because the coefficient of D in L_0 is negative, its maximum over the divisor bin uses the smallest divisor in the bin:

$$\begin{aligned} \min\big(U_{-1,\text{trunc}}\big) &= 0 - 2^{-(\alpha-1)}, \\ \max\big(L_{0,\text{trunc}}\big) &= -2^{-\beta}\delta. \end{aligned} \tag{3.40}$$

So the continuity condition becomes

$$-2^{-(\alpha-1)} \geq -2^{-\beta}\delta, \tag{3.41}$$

which is equivalent to

$$2^{-\beta}\delta \geq 2^{-(\alpha-1)}. \tag{3.42}$$

Both boundaries therefore impose the same requirement. The worst case is the smallest possible divisor index,

$$\delta_{\min} = 2^{\beta-1}.$$

Substituting it into eq. (3.39) (or equivalently eq. (3.42)) gives

$$\begin{aligned} 2^{-\beta}2^{\beta-1} &\geq 2^{-(\alpha-1)} \\ 2^{-1} &\geq 2^{-(\alpha-1)} \\ \alpha &\geq 2. \end{aligned} \tag{3.43}$$

The smallest useful choice is therefore

$$\alpha = 2, \qquad \beta = 2. \tag{3.44}$$

With these values,

$$\tau_j = \text{Int}\big(4t_j\big) \in \{-16, \ldots, 15\}, \qquad \tau_j \leq 8R_j < \tau_j + 2, \tag{3.45}$$

$$\delta = \text{Int}\big(4D_{\text{trunc}}\big) \in \{2, 3\}, \qquad \delta \leq 4D < \delta + 1. \tag{3.46}$$

At this point several concrete selector tables are possible, because the overlapping boxes may be assigned to either neighbor. The following table matches the accompanying scripts and keeps the overlap explicit. It is obtained by converting the exact boundaries in eq. (3.33) into integer inequalities using eqs. (3.45) and (3.46). For example,

$$\begin{aligned} q_{j+1} = 1: \quad & 0 \leq 2R_j \leq 2D \\ & \Longrightarrow 0 \leq 8R_j \leq 8D \\ & \Longrightarrow 0 \leq \tau_j < 2(\delta + 1), \end{aligned}$$

and similarly for the other digits. The resulting integer selector is

$$q_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 2(\delta + 1), \\ 0 & \text{if } -\delta \leq \tau_j \leq \delta - 2, \\ -1 & \text{if } -2(\delta + 1) - 2 < \tau_j \leq -2. \end{cases} \tag{3.47}$$

The inequalities are intentionally not disjoint; the shared points are the selector overlap.

3.7.2 Negative divisor

For $D \in [-1, -1/2)$, eq. (3.19) becomes

$$q_{j+1} = \begin{cases} 1 & \text{if } 2D \le 2R_j \le 0, \\ 0 & \text{if } D \le 2R_j \le -D, \\ -1 & \text{if } 0 \le 2R_j \le -2D. \end{cases} \tag{3.48}$$

The same truncation depth $\alpha = 2$ is sufficient. Now the divisor index is taken from the signed prefix of the negative divisor:

$$\delta = \mathrm{Int}\big(4D_{\mathrm{trunc}}\big) \in \{-4, -3\}, \qquad \delta \le 4D < \delta + 1. \tag{3.49}$$

Using the same integer residual index τ_j, a convenient signed selector table is

$$q_{j+1} = \begin{cases} 1 & \text{if } 2\delta - 2 < \tau_j \le -2, \\ 0 & \text{if } \delta + 1 \le \tau_j \le -(\delta + 1) - 2, \\ -1 & \text{if } 0 \le \tau_j \le -2\delta. \end{cases} \tag{3.50}$$

Again, the endpoints deliberately preserve overlap between neighboring digits.

3.8 Worked example: radix-4 division

For radix 4 we use the minimally redundant digit set

$$q_{j+1} \in \{-2, -1, 0, 1, 2\}.$$

Now $a = 2$, so

$$\rho = \frac{2}{4-1} = \frac{2}{3}. \tag{3.51}$$

3.8.1 Positive divisor

Substituting $B = 4$ and $\rho = 2/3$ into eq. (3.16) yields

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{4}{3}D \le 4R_j \le \frac{8}{3}D, \\ 1 & \text{if } \frac{1}{3}D \le 4R_j \le \frac{5}{3}D, \\ 0 & \text{if } -\frac{2}{3}D \le 4R_j \le \frac{2}{3}D, \\ -1 & \text{if } -\frac{5}{3}D \le 4R_j \le -\frac{1}{3}D, \\ -2 & \text{if } -\frac{8}{3}D \le 4R_j \le -\frac{4}{3}D. \end{cases} \tag{3.52}$$

The exact overlap between adjacent digits now has width

$$(2\rho - 1)D = \frac{1}{3}D,$$

which is narrower than in the radix-2 case and therefore forces a more precise selector.

Let the truncated residual satisfy

$$0 \le 4R_j - t_j < 2^{-(\alpha-1)}. \tag{3.53}$$

Define the integer selector variables

$$\tau_j = \mathrm{Int}\big(2^{\alpha} t_j\big), \qquad \tau_j \le 2^{\alpha+2} R_j < \tau_j + 2, \tag{3.54}$$

$$\delta = \mathrm{Int}\big(2^{\beta} D_{\mathrm{trunc}}\big), \qquad \delta \le 2^{\beta} D < \delta + 1. \tag{3.55}$$

The continuity boundary between digits $k-1$ and k is still

$$\min\big(U_{k-1,\mathrm{trunc}}\big) \ge \max\big(L_{k,\mathrm{trunc}}\big),$$

but now there are four adjacent boundaries to check: $(2,1)$, $(1,0)$, $(0,-1)$, and $(-1,-2)$. Writing them out one at a time gives

$$\text{boundary } (2,1): \quad \left(1+\frac{2}{3}\right) 2^{-\beta}\delta - 2^{-(\alpha-1)} \ge \left(2-\frac{2}{3}\right) 2^{-\beta}(\delta+1),$$

$$\frac{1}{3}2^{-\beta}\delta - \frac{4}{3}2^{-\beta} - 2^{-(\alpha-1)} \ge 0, \tag{3.56}$$

$$\text{boundary } (1,0): \quad \left(0+\frac{2}{3}\right) 2^{-\beta}\delta - 2^{-(\alpha-1)} \ge \left(1-\frac{2}{3}\right) 2^{-\beta}(\delta+1),$$

$$\frac{1}{3}2^{-\beta}\delta - \frac{1}{3}2^{-\beta} - 2^{-(\alpha-1)} \ge 0, \tag{3.57}$$

$$\text{boundary } (0,-1): \quad \left(-1+\frac{2}{3}\right) 2^{-\beta}(\delta+1) - 2^{-(\alpha-1)} \ge \left(0-\frac{2}{3}\right) 2^{-\beta}\delta,$$

$$\frac{1}{3}2^{-\beta}\delta - \frac{1}{3}2^{-\beta} - 2^{-(\alpha-1)} \ge 0, \tag{3.58}$$

$$\text{boundary } (-1,-2): \quad \left(-2+\frac{2}{3}\right) 2^{-\beta}(\delta+1) - 2^{-(\alpha-1)} \ge \left(-1-\frac{2}{3}\right) 2^{-\beta}\delta,$$

$$\frac{1}{3}2^{-\beta}\delta - \frac{4}{3}2^{-\beta} - 2^{-(\alpha-1)} \ge 0. \tag{3.59}$$

So only two distinct inequalities remain: eqs. (3.56) and (3.59) and eqs. (3.57) and (3.58). The outer boundaries are more restrictive because they subtract an extra $2^{-\beta}$ term.

A standard choice is

$$\alpha = 5, \qquad \beta = 4. \tag{3.60}$$

Then

$$\tau_j = \mathrm{Int}\big(32 t_j\big) \in \{-128, \ldots, 127\}, \qquad \tau_j \le 128 R_j < \tau_j + 2, \tag{3.61}$$

$$\delta = \mathrm{Int}\big(16 D_{\mathrm{trunc}}\big) \in \{8, 9, \ldots, 15\}, \qquad \delta \le 16 D < \delta + 1. \tag{3.62}$$

As in the radix-2 case, many equivalent tables can be formed once continuity is guaranteed. The following one matches the accompanying scripts and keeps the overlap explicit:

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\delta+1) \le \tau_j < \frac{16}{3}(\delta+1), \\ 1 & \text{if } \frac{2}{3}(\delta+1) \le \tau_j \le \frac{10}{3}\delta - 2, \\ 0 & \text{if } -\frac{4}{3}\delta \le \tau_j \le \frac{4}{3}\delta - 2, \\ -1 & \text{if } -\frac{10}{3}\delta \le \tau_j \le -\frac{2}{3}(\delta+1) - 2, \\ -2 & \text{if } -\frac{16}{3}(\delta+1) - 2 < \tau_j \le -\frac{8}{3}(\delta+1) - 2. \end{cases} \tag{3.63}$$

3.8.2 Negative divisor

For $D \in [-1, -1/2)$, eq. (3.19) gives

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}D \le 4R_j \le \frac{4}{3}D, \\ 1 & \text{if } \frac{5}{3}D \le 4R_j \le \frac{1}{3}D, \\ 0 & \text{if } \frac{2}{3}D \le 4R_j \le -\frac{2}{3}D, \\ -1 & \text{if } -\frac{1}{3}D \le 4R_j \le -\frac{5}{3}D, \\ -2 & \text{if } -\frac{4}{3}D \le 4R_j \le -\frac{8}{3}D. \end{cases} \tag{3.64}$$

The same precision pair $(\alpha, \beta) = (5, 4)$ is sufficient. The signed divisor index is now

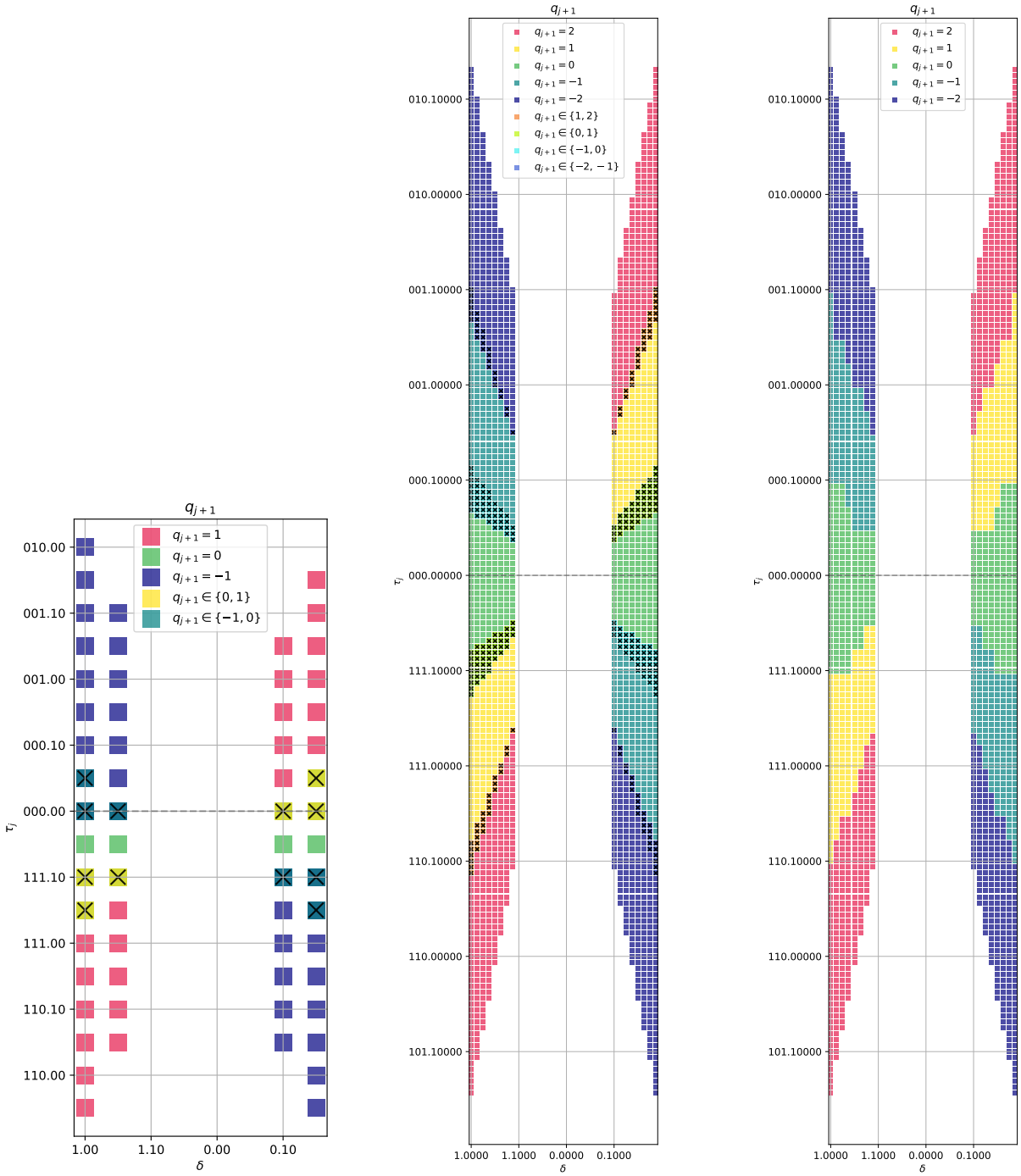
$$\delta = \mathrm{Int}\big(16D_{\mathrm{trunc}}\big) \in \{-16, -15, \ldots, -9\}, \qquad \delta \le 16D < \delta + 1. \tag{3.65}$$

A matching signed selector table is

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{16}{3}\delta - 2 < \tau_j \le \frac{8}{3}\delta - 2, \\ 1 & \text{if } \frac{10}{3}(\delta + 1) \le \tau_j \le \frac{2}{3}\delta - 2, \\ 0 & \text{if } \frac{4}{3}(\delta + 1) \le \tau_j \le -\frac{4}{3}(\delta + 1) - 2, \\ -1 & \text{if } -\frac{2}{3}\delta \le \tau_j \le -\frac{10}{3}(\delta + 1) - 2, \\ -2 & \text{if } -\frac{8}{3}\delta \le \tau_j \le -\frac{16}{3}\delta. \end{cases} \tag{3.66}$$

3.9 Selector plots

Fig. 1 visualizes the quotient-digit regions produced by the basic and optimized selectors. The basic plots leave overlap explicit; the optimized plot shows one deterministic way to assign the overlap boxes without changing correctness. The optimization itself is described later in section 5.



(a) Radix-2 basic QDS **(b)** Radix-4 basic QDS **(c)** Radix-4 optimized QDS

Figure 1: Quotient-digit selection regions. The horizontal axis is the truncated divisor index δ, and the vertical axis is the truncated residual index τ_j.

4 Square Root by Digit Recurrence

Square root follows the same overall plan as division, but the digit-selection bounds depend on the current partial root as well as on the residual. That dependence is what makes square-root selectors slightly more intricate. We begin from the exact relation

$$X = S_n^2 + r, \tag{4.1}$$

where X is the radicand, S_n the final root, and r the final *unscaled* remainder.

4.1 Root expansion

Let S_j denote the partial root after j selected digits. The next digit update is

$$S_{j+1} = S_j + s_{j+1}B^{-(j+1)}. \tag{4.2}$$

Repeated substitution gives

$$S_{j+1} = S_0 + \sum_{i=0}^{j} s_{i+1}B^{-(i+1)}, \tag{4.3}$$

and therefore

$$S_n = S_0 + \sum_{i=0}^{n-1} s_{i+1}B^{-(i+1)}. \tag{4.4}$$

4.2 Approximation error and containment

Define the root error

$$\varepsilon_{j+1} = \sqrt{X} - S_{j+1}. \tag{4.5}$$

As in division, the tail of the digit series bounds this error:

$$\varepsilon_{j+1} = \sum_{i=j+1}^{\infty} s_{i+1}B^{-(i+1)}. \tag{4.6}$$

Since $|s_{i+1}| \le a$,

$$\begin{aligned}
|\varepsilon_{j+1}| &= \left| \sum_{i=j+1}^{\infty} s_{i+1}B^{-(i+1)} \right| \\
&\le \sum_{i=j+1}^{\infty} aB^{-(i+1)} \\
&= aB^{-(j+2)} \sum_{k=0}^{\infty} B^{-k} \\
&= aB^{-(j+2)} \cdot \frac{1}{1-B^{-1}} \\
&= \frac{a}{B-1} B^{-(j+1)} \\
&= \rho B^{-(j+1)}.
\end{aligned} \tag{4.7}$$

So

$$\left|\sqrt{X} - S_{j+1}\right| \le \rho B^{-(j+1)}. \tag{4.8}$$

Expanding the absolute-value inequality gives

$$S_{j+1} - \rho B^{-(j+1)} \le \sqrt{X} \le S_{j+1} + \rho B^{-(j+1)}. \tag{4.9}$$

Now square all three terms:

$$\begin{aligned}\left(S_{j+1} - \rho B^{-(j+1)}\right)^2 \le X \le \left(S_{j+1} + \rho B^{-(j+1)}\right)^2 \\ S_{j+1}^2 - 2\rho S_{j+1} B^{-(j+1)} + \rho^2 B^{-2(j+1)} \le X \le S_{j+1}^2 + 2\rho S_{j+1} B^{-(j+1)} + \rho^2 B^{-2(j+1)}.\end{aligned} \tag{4.10}$$

Subtract S_{j+1}^2 from all three parts:

$$\rho^2 B^{-2(j+1)} - 2\rho S_{j+1} B^{-(j+1)} \le X - S_{j+1}^2 \le \rho^2 B^{-2(j+1)} + 2\rho S_{j+1} B^{-(j+1)}. \tag{4.11}$$

Define the scaled residual

$$R_{j+1} = B^{j+1}(X - S_{j+1}^2). \tag{4.12}$$

Multiplying eq. (4.11) by B^{j+1} gives the containment condition

$$\rho^2 B^{-(j+1)} - 2\rho S_{j+1} \le R_{j+1} \le \rho^2 B^{-(j+1)} + 2\rho S_{j+1}. \tag{4.13}$$

At the last step,

$$R_n = B^n(X - S_n^2) = B^n r. \tag{4.14}$$

4.3 Residual recurrence

Starting from eq. (4.12) and substituting the update eq. (4.2),

$$\begin{aligned}R_{j+1} &= B^{j+1}(X - S_{j+1}^2) \\ &= B^{j+1}\left(X - \left(S_j + s_{j+1} B^{-(j+1)}\right)^2\right) \\ &= B^{j+1}(X - S_j^2) - 2S_j s_{j+1} - s_{j+1}^2 B^{-(j+1)} \\ &= BR_j - s_{j+1}\left(2S_j + s_{j+1} B^{-(j+1)}\right).\end{aligned} \tag{4.15}$$

Compared with division, the update now involves the current partial root S_j and the square of the new digit. That dependence is the source of the additional selector complexity.

4.4 Ideal digit-selection intervals

Substitute eq. (4.15) into the containment inequality eq. (4.13):

$$\begin{aligned}\rho^2 B^{-(j+1)} - 2\rho S_{j+1} &\le BR_j - s_{j+1}\left(2S_j + s_{j+1} B^{-(j+1)}\right) \\ &\le \rho^2 B^{-(j+1)} + 2\rho S_{j+1}.\end{aligned} \tag{4.16}$$

Next replace S_{j+1} by $S_j + s_{j+1} B^{-(j+1)}$:

$$\begin{aligned}\rho^2 B^{-(j+1)} - 2\rho\left(S_j + s_{j+1} B^{-(j+1)}\right) &\le BR_j - s_{j+1}\left(2S_j + s_{j+1} B^{-(j+1)}\right) \\ &\le \rho^2 B^{-(j+1)} + 2\rho\left(S_j + s_{j+1} B^{-(j+1)}\right).\end{aligned} \tag{4.17}$$

Now move the digit-dependent term on the right-hand side of the residual recurrence to the outside. For the left inequality,

$$\begin{aligned} BR_j &\geq s_{j+1}\left(2S_j + s_{j+1}B^{-(j+1)}\right) + \rho^2 B^{-(j+1)} - 2\rho\left(S_j + s_{j+1}B^{-(j+1)}\right) \\ &= 2(s_{j+1}-\rho)S_j + \left(s_{j+1}^2 - 2\rho s_{j+1} + \rho^2\right)B^{-(j+1)} \\ &= 2(s_{j+1}-\rho)S_j + (s_{j+1}-\rho)^2 B^{-(j+1)}. \end{aligned}$$

Similarly, for the right inequality,

$$\begin{aligned} BR_j &\leq s_{j+1}\left(2S_j + s_{j+1}B^{-(j+1)}\right) + \rho^2 B^{-(j+1)} + 2\rho\left(S_j + s_{j+1}B^{-(j+1)}\right) \\ &= 2(s_{j+1}+\rho)S_j + (s_{j+1}+\rho)^2 B^{-(j+1)}. \end{aligned}$$

Therefore the exact selector interval is

$$2(s_{j+1}-\rho)S_j + (s_{j+1}-\rho)^2 B^{-(j+1)} \leq BR_j \leq 2(s_{j+1}+\rho)S_j + (s_{j+1}+\rho)^2 B^{-(j+1)}. \tag{4.18}$$

Unlike division, the bounds depend on S_j itself and on an explicit $B^{-(j+1)}$ term. This is why the selector can be iteration-dependent during startup.

4.5 Initialization

With the tutorial normalization $X \in [1/4, 1)$, the exact root satisfies

$$\sqrt{X} \in \left[\frac{1}{2}, 1\right). \tag{4.19}$$

That suggests a convenient starting point from above:

$$S_0 = 1. \tag{4.20}$$

The first update is

$$S_1 = S_0 + s_1 B^{-1} = 1 + s_1 B^{-1}. \tag{4.21}$$

Applying the general error bound eq. (4.8) with $j = 0$ gives

$$\left|\sqrt{X} - S_1\right| \leq \rho B^{-1}, \tag{4.22}$$

so

$$S_1 - \rho B^{-1} \leq \sqrt{X} \leq S_1 + \rho B^{-1}. \tag{4.23}$$

Since $S_1 = 1 + s_1 B^{-1}$, digit s_1 covers the interval

$$1 + (s_1 - \rho)\, B^{-1} \leq \sqrt{X} \leq 1 + (s_1 + \rho)\, B^{-1}. \tag{4.24}$$

Because the true root lies in $[1/2, 1)$, we may restrict the first digit to the nonpositive half of the digit set:

$$s_1 \in \{0, -1, \ldots, -a\}.$$

That is exactly why the chosen normalization is convenient.

It is easy to verify coverage of $[1/2, 1)$. The rightmost possible first-step interval is obtained from $s_1 = 0$:

$$1 - \rho B^{-1} \leq \sqrt{X} \leq 1 + \rho B^{-1}.$$

The leftmost interval is obtained from $s_1 = -a$:

$$1 - (a+\rho)B^{-1} \le \sqrt{X} \le 1 - (a-\rho)B^{-1}.$$

Since $a = \rho(B-1)$, the left endpoint becomes

$$1 - (a+\rho)B^{-1} = 1 - \rho.$$

Because $\rho > 1/2$, the left edge reaches below $1/2$, and the first digit always exists.

For the two radices used later, the startup intervals are especially simple.

Radix-2. With $B = 2$ and $\rho = 1$,

$$\begin{aligned} s_1 = 0 &\Longrightarrow \frac{1}{2} \le \sqrt{X} \le \frac{3}{2}, \\ s_1 = -1 &\Longrightarrow 0 \le \sqrt{X} \le 1. \end{aligned}$$

Squaring yields

$$\begin{aligned} s_1 = 0 &\Longrightarrow \frac{1}{4} \le X \le \frac{9}{4}, \\ s_1 = -1 &\Longrightarrow 0 \le X \le 1. \end{aligned}$$

Radix-4. With $B = 4$ and $\rho = 2/3$,

$$\begin{aligned} s_1 = 0 &\Longrightarrow \frac{5}{6} \le \sqrt{X} \le \frac{7}{6}, \\ s_1 = -1 &\Longrightarrow \frac{7}{12} \le \sqrt{X} \le \frac{11}{12}, \\ s_1 = -2 &\Longrightarrow \frac{1}{3} \le \sqrt{X} \le \frac{2}{3}. \end{aligned}$$

Squaring yields

$$\begin{aligned} s_1 = 0 &\Longrightarrow \frac{25}{36} \le X \le \frac{49}{36}, \\ s_1 = -1 &\Longrightarrow \frac{49}{144} \le X \le \frac{121}{144}, \\ s_1 = -2 &\Longrightarrow \frac{1}{9} \le X \le \frac{4}{9}. \end{aligned}$$

4.6 From exact intervals to a practical root selector

The root selector again works with truncated quantities. In the square-root case, however, the additive $B^{-(j+1)}$ terms in eq. (4.18) are still large enough during the earliest iterations that the startup selector and the steady-state selector need not coincide. Let the retained residual prefix satisfy

$$0 \le BR_j - t_j < 2^{-(\alpha-1)}. \tag{4.25}$$

The partial root must also be truncated. Write its truncated integer index as σ_j so that

$$\sigma_j \le 2^\beta S_j \le \sigma_j + 1 - 2^\beta \mathrm{ulp}, \tag{4.26}$$

where the final term reminds us that S_j itself is already quantized to the current root ulp. The generic selector-design task is therefore similar to the division case, except that the exact interval boundaries now depend on both S_j and the iteration index j. In practice one checks the continuity condition at each adjacent digit boundary and, if the required overlap is not yet present in the earliest iterations, introduces special startup paths. The worked examples below show exactly how that happens.

4.7 Worked example: radix-2 square root

For radix 2 we use the digit set

$$s_{j+1} \in \{-1, 0, 1\},$$

so again $a = 1$ and $\rho = 1$. Substituting these values into eq. (4.18) gives the exact selector intervals

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \le 2R_j \le 4S_j + 4 \cdot 2^{-(j+1)}, \\ 0 & \text{if } -2S_j + 2^{-(j+1)} \le 2R_j \le 2S_j + 2^{-(j+1)}, \\ -1 & \text{if } -4S_j + 4 \cdot 2^{-(j+1)} \le 2R_j \le 0. \end{cases} \tag{4.27}$$

Now define the truncated integer variables

$$\tau_j = \mathrm{Int}\big(2^{\alpha} t_j\big), \qquad \tau_j \le 2^{\alpha+1} R_j < \tau_j + 2, \tag{4.28}$$

$$\sigma_j = \mathrm{Int}\big(2^{\beta} S_{j,\mathrm{trunc}}\big), \qquad \sigma_j \le 2^{\beta} S_j \le \sigma_j + 1 - 2^{\beta}\mathrm{ulp}. \tag{4.29}$$

Since $S_j \in [1/2, 1]$, the partial-root index ranges over

$$\sigma_j \in \{2^{\beta-1}, 2^{\beta-1} + 1, \ldots, 2^{\beta} - 1, 2^{\beta}\}. \tag{4.30}$$

For the interior steady-state selector we choose

$$\alpha = 1, \qquad \beta = 2. \tag{4.31}$$

At this precision the full truncated-root range is $\sigma_j \in \{2, 3, 4\}$. The interior state is $\sigma_j = 3$; the boundary values $\sigma_j = 2$ and $\sigma_j = 4$ correspond to startup paths near $X = 1/4$ and $X = 1$, and are treated separately below. Under that convention,

$$\tau_j = \mathrm{Int}\big(2t_j\big) \in \{-16, \ldots, 15\}, \qquad \tau_j \le 4R_j < \tau_j + 2, \tag{4.32}$$

$$\sigma_j = \mathrm{Int}\big(4S_{j,\mathrm{trunc}}\big) \in \{3\}, \qquad \sigma_j \le 4S_j \le \sigma_j + 1 - 4 \cdot 2^{-j}. \tag{4.33}$$

The continuity condition now contains explicit j-dependent terms. Writing the two adjacent boundaries explicitly produces

$$\text{boundary } (1, 0): \quad 2 \cdot 2^{-\beta}\sigma_j - 2^{-(\alpha-1)} \ge 0, \tag{4.34}$$

$$\text{boundary } (0, -1): \quad 2 \cdot 2^{-\beta}\sigma_j - 2^{-(\alpha-1)} - B^{-(j+1)} \ge 0. \tag{4.35}$$

The last one is the most restrictive. Using $B = 2$, $\alpha = 1$, $\beta = 2$, and the worst-case *interior* partial-root index $\sigma_j = 3$, it becomes

$$\begin{aligned} 2 \cdot 2^{-2} \cdot 3 - 2^{-0} - 2^{-(j+1)} &\ge 0 \\ \frac{1}{2} - 2^{-(j+1)} &\ge 0. \end{aligned} \tag{4.36}$$

Thus the interior radix-2 selector is valid for every post-initialization iteration $j \geq 0$. The only remaining cases are the two boundary startup paths omitted from the interior state $\sigma_j = 3$.

For $j \geq 0$, a convenient integer selector table is

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 2(\sigma_j + 1), \\ 0 & \text{if } -(\sigma_j - 1) \leq \tau_j \leq \sigma_j - 2, \\ -1 & \text{if } -2(\sigma_j + 1) - 2 < \tau_j \leq -2. \end{cases} \tag{4.37}$$

4.7.1 Startup path: radicand close to $1/4$

When the radicand is near the low end of the normalized interval, initialization chooses

$$s_1 = -1.$$

This is exactly the boundary case $\sigma_j = 2$ that was excluded from the interior selector above. Suppose the following digits remain zero:

$$s_2 = s_3 = \ldots = s_j = 0.$$

Then

$$S_1 = S_0 + s_1 2^{-1} = 1 - \frac{1}{2} = \frac{1}{2}, \tag{4.38}$$

and hence

$$S_j = \frac{1}{2}, \qquad j \geq 1. \tag{4.39}$$

Therefore

$$\sigma_j = \mathrm{Int}(4 \cdot 0.10) = 2. \tag{4.40}$$

The first residual is

$$R_1 = 2\left(X - \frac{1}{4}\right) \geq 0, \tag{4.41}$$

and if all later digits remain zero, the recurrence reduces to

$$R_k = 2R_{k-1}, \qquad k \geq 2. \tag{4.42}$$

Thus the residual remains nonnegative on this path. Since truncation is one-sided,

$$0 \leq 2R_j - t_j < 2^{-(\alpha-1)} \quad \Longrightarrow \quad t_j > -2^{-(\alpha-1)}.$$

With $\alpha = 1$, this implies

$$\tau_j = \mathrm{Int}(2t_j) \geq -1. \tag{4.43}$$

Therefore the negative-digit region of eq. (4.37) is unreachable, and the selector reduces to

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 2(\sigma_j + 1), \\ 0 & \text{if } -1 \leq \tau_j \leq \sigma_j - 2, \end{cases} \qquad \sigma_j = 2. \tag{4.44}$$

4.7.2 Startup path: radicand close to 1

At the high end of the normalized interval, initialization may select a run of zeros:

$$s_1 = s_2 = \ldots = s_j = 0.$$

This is the complementary boundary case $\sigma_j = 4$. Then the partial root remains fixed at

$$S_j = S_0 = 1, \tag{4.45}$$

so

$$\sigma_j = \mathrm{Int}(4 \cdot 1.00) = 4. \tag{4.46}$$

Because $X < 1$,

$$R_1 = 2(X - 1) < 0. \tag{4.47}$$

If subsequent digits are zero, then again $R_k = 2R_{k-1}$, so the residual stays negative. Consequently the truncated residual estimate also stays negative:

$$\tau_j \leq -1. \tag{4.48}$$

Therefore the positive-digit region of eq. (4.37) is unreachable, and the selector reduces to

$$s_{j+1} = \begin{cases} 0 & \text{if } -(\sigma_j - 1) \leq \tau_j \leq -1, \\ -1 & \text{if } -2\sigma_j - 2 < \tau_j \leq -2, \end{cases} \qquad \sigma_j = 4. \tag{4.49}$$

4.8 Worked example: radix-4 square root

For radix 4 we use the digit set

$$s_{j+1} \in \{-2, -1, 0, 1, 2\},$$

so again $a = 2$ and $\rho = 2/3$. Substituting these values into eq. (4.18) gives the exact selector intervals

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}S_j + \frac{16}{9}4^{-(j+1)} \leq 4R_j \leq \frac{16}{3}S_j + \frac{64}{9}4^{-(j+1)}, \\ 1 & \text{if } \frac{2}{3}S_j + \frac{1}{9}4^{-(j+1)} \leq 4R_j \leq \frac{10}{3}S_j + \frac{25}{9}4^{-(j+1)}, \\ 0 & \text{if } -\frac{4}{3}S_j + \frac{4}{9}4^{-(j+1)} \leq 4R_j \leq \frac{4}{3}S_j + \frac{4}{9}4^{-(j+1)}, \\ -1 & \text{if } -\frac{10}{3}S_j + \frac{25}{9}4^{-(j+1)} \leq 4R_j \leq -\frac{2}{3}S_j + \frac{1}{9}4^{-(j+1)}, \\ -2 & \text{if } -\frac{16}{3}S_j + \frac{64}{9}4^{-(j+1)} \leq 4R_j \leq -\frac{8}{3}S_j + \frac{16}{9}4^{-(j+1)}. \end{cases} \tag{4.50}$$

Now define the truncated integer variables

$$\tau_j = \mathrm{Int}\big(2^{\alpha} t_j\big), \qquad \tau_j \leq 2^{\alpha+2} R_j < \tau_j + 2, \tag{4.51}$$

$$\sigma_j = \mathrm{Int}\big(2^{\beta} S_{j,\mathrm{trunc}}\big), \qquad \sigma_j \leq 2^{\beta} S_j \leq \sigma_j + 1 - 2^{\beta}\mathrm{ulp}. \tag{4.52}$$

For the interior steady-state selector we choose

$$\alpha = 4, \qquad \beta = 4. \tag{4.53}$$

At this precision the full truncated-root range is $\sigma_j \in \{8, 9, \ldots, 15, 16\}$. The interior states are $\sigma_j \in \{9, \ldots, 15\}$; the boundary values $\sigma_j = 8$ and $\sigma_j = 16$ correspond to startup paths near $X = 1/4$ and $X = 1$, and are treated separately below. Under that convention,

$$\tau_j = \mathrm{Int}\big(16 t_j\big) \in \{-128, \ldots, 127\}, \qquad \tau_j \leq 64 R_j < \tau_j + 2, \tag{4.54}$$

$$\sigma_j = \mathrm{Int}\big(16 S_{j,\mathrm{trunc}}\big) \in \{9, \ldots, 15\}, \qquad \sigma_j \leq 16 S_j \leq \sigma_j + 1 - 16 \cdot 4^{-j}. \tag{4.55}$$

The continuity condition now contains explicit j-dependent terms. Writing the four adjacent boundaries explicitly produces

$$\text{boundary } (2,1): \quad \frac{2}{3}2^{-\beta}\sigma_j - \frac{8}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0, \tag{4.56}$$

$$\text{boundary } (1,0): \quad \frac{2}{3}2^{-\beta}\sigma_j - \frac{2}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0, \tag{4.57}$$

$$\text{boundary } (0,-1): \quad \frac{2}{3}2^{-\beta}\sigma_j - \frac{2}{3}2^{-\beta} - 2^{-(\alpha-1)} - \frac{4}{9}B^{-(j+1)} \geq 0, \tag{4.58}$$

$$\text{boundary } (-1,-2): \quad \frac{2}{3}2^{-\beta}\sigma_j - \frac{8}{3}2^{-\beta} - 2^{-(\alpha-1)} - \frac{25}{9}B^{-(j+1)} \geq 0. \tag{4.59}$$

The last one is the most restrictive. Using $B = 4$, $\alpha = 4$, $\beta = 4$, and the worst-case *interior* partial-root index $\sigma_j = 9$, it becomes

$$\begin{aligned} \frac{2}{3} \cdot 2^{-4} \cdot 9 - \frac{8}{3} \cdot 2^{-4} - 2^{-3} - \frac{25}{9}4^{-(j+1)} &\geq 0 \\ \frac{1}{12} - \frac{25}{9}4^{-(j+1)} &\geq 0. \end{aligned} \tag{4.60}$$

Thus the interior radix-4 selector is a steady-state selector: it applies for $j \geq 2$ and for interior partial-root indices $\sigma_j \in \{9, \ldots, 15\}$. The remaining work is to cover the two boundary startup paths.

For $j \geq 2$, a convenient integer selector table is

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\sigma_j + 1) \leq \tau_j < \frac{16}{3}(\sigma_j + 1), \\ 1 & \text{if } \frac{2}{3}(\sigma_j + 1) \leq \tau_j \leq \frac{10}{3}\sigma_j - 2, \\ 0 & \text{if } -\frac{4}{3}\left(\sigma_j - \frac{1}{12}\right) \leq \tau_j \leq \frac{4}{3}\sigma_j - 2, \\ -1 & \text{if } -\frac{10}{3}\left(\sigma_j - \frac{5}{24}\right) \leq \tau_j \leq -\frac{2}{3}(\sigma_j + 1) - 2, \\ -2 & \text{if } -\frac{16}{3}(\sigma_j + 1) - 2 < \tau_j \leq -\frac{8}{3}(\sigma_j + 1) - 2. \end{cases} \tag{4.61}$$

4.8.1 Startup path: radicand close to $1/4$

When the radicand is close to $1/4$, initialization chooses the most negative first digit,

$$s_1 = -2,$$

which is the boundary case $\sigma_j = 8$ omitted from the interior selector. If the next digits are then zero,

$$S_1 = 1 - \frac{2}{4} = \frac{1}{2}, \tag{4.62}$$

so

$$S_j = \frac{1}{2}, \qquad j \geq 1. \tag{4.63}$$

Hence

$$\sigma_j = \mathrm{Int}(16 \cdot 0.1000) = 8. \tag{4.64}$$

The residual is nonnegative on this path because

$$R_1 = 4\left(X - \frac{1}{4}\right) \geq 0, \tag{4.65}$$

and with zero digits thereafter the recurrence reduces to $R_k = 4R_{k-1}$. Since truncation is one-sided,

$$0 \le 4R_j - t_j < 2^{-(\alpha-1)} \quad \Longrightarrow \quad t_j > -2^{-(\alpha-1)}.$$

For $\alpha = 4$ this implies

$$\tau_j = \mathrm{Int}(16t_j) \ge -1. \tag{4.66}$$

Since the negative selector regions are unreachable, the table reduces to

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\sigma_j + 1) \le \tau_j < \frac{16}{3}(\sigma_j + 1), \\ 1 & \text{if } \frac{2}{3}(\sigma_j + 1) \le \tau_j \le \frac{10}{3}\sigma_j - 2, \\ 0 & \text{if } -1 \le \tau_j \le \frac{4}{3}\sigma_j - 2, \end{cases} \qquad \sigma_j = 8. \tag{4.67}$$

4.8.2 Startup path: radicand close to 1

When the radicand is close to 1, the selected digits can remain zero for several steps:

$$s_1 = s_2 = \ldots = s_j = 0.$$

This is the complementary boundary case $\sigma_j = 16$. Then

$$S_j = S_0 = 1, \tag{4.68}$$

so

$$\sigma_j = \mathrm{Int}(16 \cdot 1.0000) = 16. \tag{4.69}$$

The residual stays negative because

$$R_1 = 4(X - 1) < 0, \tag{4.70}$$

and if the subsequent digits are zero then again $R_k = 4R_{k-1}$. Thus the truncated residual index satisfies

$$\tau_j \le -1. \tag{4.71}$$

Now revisit the most restrictive continuity inequality eq. (4.59) with $\sigma_j = 16$:

$$\begin{aligned} \frac{2}{3} \cdot 2^{-4} \cdot 16 - \frac{8}{3} \cdot 2^{-4} - 2^{-3} - \frac{25}{9} 4^{-(j+1)} &\ge 0 \\ \frac{3}{8} - \frac{25}{9} 4^{-(j+1)} &\ge 0. \end{aligned} \tag{4.72}$$

This fails only at $j = 0$, which is already handled by initialization, and holds for $j = 1$ because

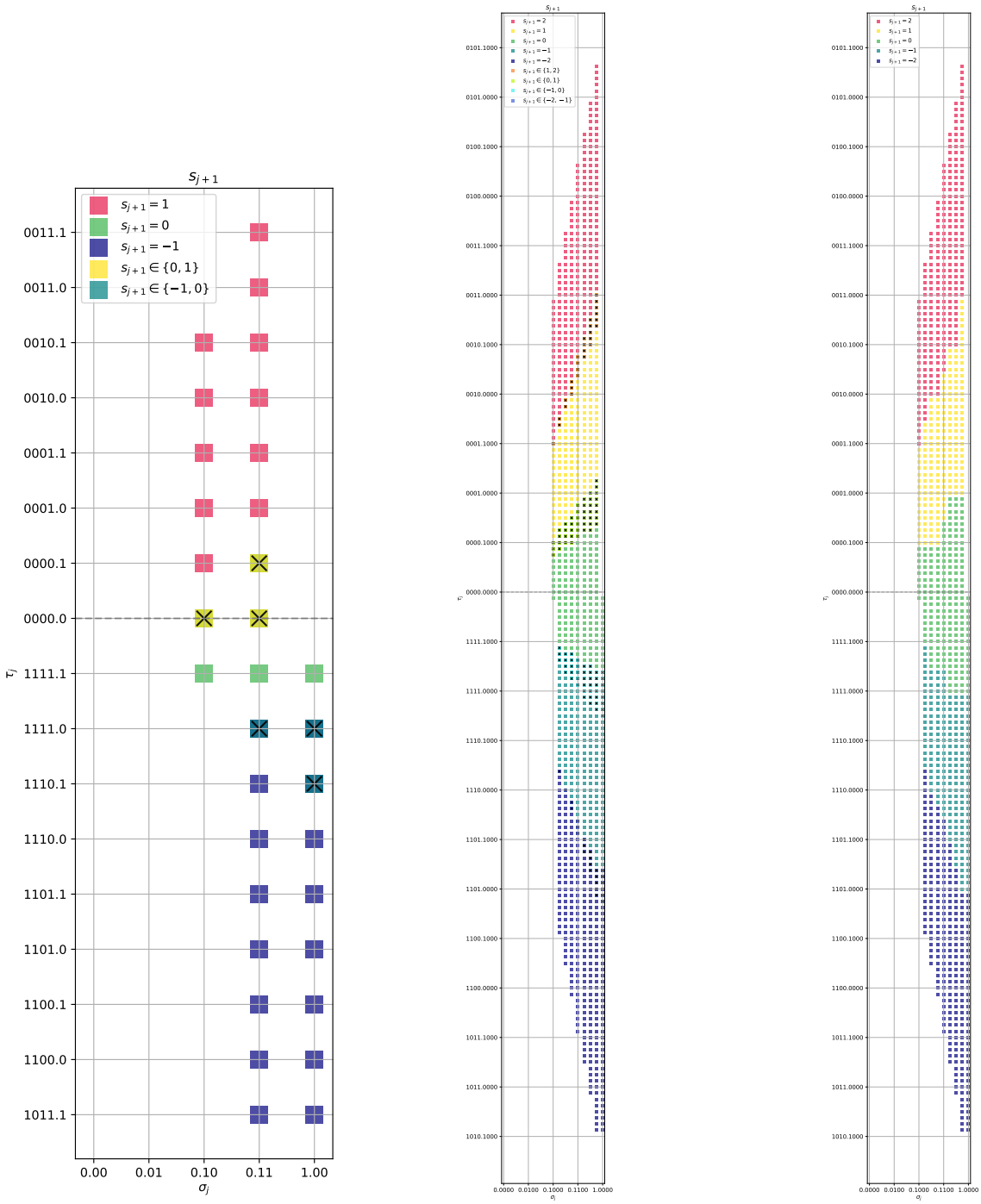
$$\frac{3}{8} - \frac{25}{144} = \frac{29}{144} > 0.$$

So along the near-one startup path, the steady-state precision is already sufficient from the first post-initialization iteration onward. Since the positive selector regions are unreachable, the table reduces to

$$s_{j+1} = \begin{cases} 0 & \text{if } -\frac{4}{3}\left(\sigma_j - \frac{1}{3}\right) \le \tau_j \le -1, \\ -1 & \text{if } -\frac{10}{3}\left(\sigma_j - \frac{5}{6}\right) \le \tau_j \le -\frac{2}{3}\sigma_j - 2, \\ -2 & \text{if } -\frac{16}{3}\sigma_j - 2 < \tau_j \le -\frac{8}{3}\sigma_j - 2, \end{cases} \qquad \sigma_j = 16. \tag{4.73}$$

4.9 Selector plots

Fig. 2 shows the root-digit regions for the radix-2 and radix-4 selectors. As in the division plots, the basic versions keep the overlap regions explicit, while the optimized plot shows one deterministic overlap assignment.



(a) Radix-2 basic RDS **(b)** Radix-4 basic RDS **(c)** Radix-4 optimized RDS

Figure 2: Root-digit selection regions. The horizontal axis is the truncated partial-root index σ_j, and the vertical axis is the truncated residual index τ_j.

5 Resolving Overlap Deterministically

The "basic" selectors in sections 3 and 4 intentionally leave overlap visible. That is mathematically natural: inside an overlap box, more than one digit is valid, so the selector really does have freedom. A hardware implementation, however, usually wants a deterministic answer. The practical question is therefore not whether overlap exists, but how to assign those overlap boxes in a way that preserves correctness and keeps the logic simple.

5.1 What is being optimized?

Fix one column of the selector table, i.e. fix one truncated divisor index δ for division or one truncated partial-root index σ_j for square root. Within that column, neighboring digits have integer residual ranges that overlap. Suppose the lower digit is valid up to an integer threshold x, and the higher digit is valid from an integer threshold y onward, with

$$y \leq x.$$

Then the inclusive interval $[y, x]$ is the overlap region. To eliminate the overlap, we need to replace the pair (x, y) by a new pair (x', y') such that

$$x' < y' \qquad \text{and typically} \qquad y' = x' + 1. \tag{5.1}$$

The lower digit then occupies all indices up to x', and the higher digit occupies all indices from y' onward.

Many such choices are valid. The criterion used by the accompanying scripts is not mathematical necessity but logic simplicity: choose x' and y' so that the boundaries align with convenient congruence classes, which in turn simplifies Karnaugh-map grouping and downstream logic minimization.

5.2 A simple modulo-aligned transformation

One practical rule is the following transformation on the overlap endpoints. Let

$$\text{diff} = x - y. \tag{5.2}$$

The value of `diff` indicates how wide the overlap is. The transformation chooses a modulus M based on that width:

$$M = \begin{cases} 4 & \text{if } 1 \leq \text{diff} \leq 2, \\ 8 & \text{if } 3 \leq \text{diff} \leq 6, \\ 16 & \text{if } 7 \leq \text{diff} \leq 14. \end{cases} \tag{5.3}$$

The degenerate case $\text{diff} = 0$ is handled separately.

For $\text{diff} > 0$, define

$$\text{base} = \left\lfloor \frac{y-1}{M} \right\rfloor M, \tag{5.4}$$

$$\text{threshold} = \text{base} + M - 1. \tag{5.5}$$

Then there are two candidate no-overlap pairs:

$$(x'_{\text{low}}, y'_{\text{low}}) = \left(\text{base} + \frac{M}{2} - 1,\ \text{base} + \frac{M}{2}\right), \tag{5.6}$$

$$(x'_{\text{high}}, y'_{\text{high}}) = (\text{base} + M - 1,\ \text{base} + M). \tag{5.7}$$

The rule is then

$$(x', y') = \begin{cases} (x'_{\text{low}}, y'_{\text{low}}) & \text{if } x < \text{threshold}, \\ (x'_{\text{high}}, y'_{\text{high}}) & \text{if } x \geq \text{threshold}. \end{cases} \tag{5.8}$$

When $\text{diff} = 0$, the overlap consists of a single point. The scripts break that tie by parity:

$$(x', y') = \begin{cases} (x, x+1) & \text{if } x \text{ is odd}, \\ (x-1, x) & \text{if } x \text{ is even}. \end{cases} \tag{5.9}$$

Either choice removes the overlap; the parity rule simply keeps the boundaries aligned on a regular grid.

5.3 Why this helps

The point of eqs. (5.8) and (5.9) is not that it is the only valid transformation. Its value is that the new boundaries fall into a small number of repeated patterns:

- mod-4 boundaries for narrow overlaps,
- mod-8 boundaries for medium overlaps, and
- mod-16 boundaries for wide overlaps.

Those regular patterns are easier to minimize than an arbitrary assignment of every overlap point to its left or right neighbor. This is exactly what distinguishes the "basic" and "optimized" plots in figs. 1 and 2. The basic plots expose every valid overlap region; the optimized plots apply a deterministic modulo-aligned assignment to remove those overlaps.

6 Conclusion

Digit-recurrence algorithms solve a hardware problem by exploiting a mathematical one. The mathematical requirement is to keep the residual bounded while producing one output digit per iteration. The hardware requirement is to make each step of the loop fast enough for a practical divider or square-root unit. Redundancy connects the two: once adjacent digit intervals overlap, truncated state is sufficient for selection, so the critical path no longer needs an exact full-width carry-propagating comparison.

This tutorial developed that idea from first principles. For division we started from the exact quotient relation, derived the digit expansion, bounded the tail of the redundant representation, introduced the scaled residual, and obtained the recurrence

$$R_{j+1} = BR_j - q_{j+1}D.$$

That recurrence led directly to exact digit-selection intervals and then to practical radix-2 and radix-4 selector tables built from truncated residual and divisor information.

For square root the same structure remains, but the recurrence

$$R_{j+1} = BR_j - s_{j+1}\left(2S_j + s_{j+1}B^{-(j+1)}\right)$$

also depends on the evolving partial root. That extra dependence explains why square-root startup requires more care and why it is useful to distinguish interior steady-state selectors from boundary startup paths.

Two implementation lessons matter most. First, the continuity condition is the bridge from theory to hardware: it quantifies exactly how much truncation the selector can tolerate while still leaving at least one valid digit in every table entry. Second, once continuity is guaranteed, overlap becomes a design resource. It may be shown explicitly in a pedagogical "basic" selector, or resolved deterministically to produce a smaller and easier-to-minimize implementation.

The digit-recurrence framework is therefore more than a recurrence formula. It provides a full workflow for choosing the digit set, normalizing the operands, bounding the residual, sizing the selector inputs, and converting overlap into real hardware logic. That combination is why SRT-style division and square root remain central building blocks in floating-point units.